



Formal

Assessment Task.

MEMORANDUM

Apprentice Level 1 Duration: 45 minutes Lesson Numbers: 11 to 20

Grade/Class:

Questions	1	2	3	4	5
Blooms	Remember	Understand	Apply	Analyse	Evaluate/ Create
Pillars	(Coding/ Robotics/ Digital Concepts)	(Coding/ Robotics/ Digital Concepts)	(Coding/ Robotics/ Digital Concepts)	(Coding/ Robotics/ Digital Concepts)	(Coding/ Robotics/ Digital Concepts)
Learners Mark					
Total	5	5	11	14	10
				TOTAL	
					45

Please read the following instructions carefully.

- This paper consists of 10 pages and 5 questions.
- Answers must be completed with a blue pen.
- Read each question carefully.
- Pay attention to mark allocation.



Section A – Knowledge: Recall (Remembering) (CAPS Pillar)

Question 1: Multiple choice

Choose the correct answer from the four possible answers to each question below. Write your answer in the answer box provided.

1.1. Which one of the blocks allows your sprite to interact by touching colours? (1) A.

Control

B. Sound

C. Sensing

D. Events

1.2. Choose the answer that best describes a variable: (1) A. Allows you to move sprites across the screen.

B. Values that can change.

C. The most important coding block in Scratch.

D. A way to make decisions in programming.



1.3. Which of the following will you use to multiply two variables? (1) A.

B.

C.

D.

2



1.4. Micro:bits should **NOT** be handled in this way... (1) A. Work on a smooth, flat surface.

B. Store your MicroBit in its case when you are not using it.

C. Only use the correct battery and charger for your MicroBit.

D. Leave the MicroBit in the sun all day to charge.

1.5. What does the acronym LED stand for? (1) A. Light-Emitting Diode

B. Lidar-Enabled Drone

C. Long Emergency Drill

Total [5]**Section B – Explain concepts or Ideas (Understanding) (CAPS Pillar)****Question 2:** Match Columns A and B and write your answers in Column C.

Column A	Column B	Column C (Eg 2.6F)
2.1. Variable	A. A set of rules and instructions to follow.	2.1 B
2.2. Pen up block	B. Used to keep score in a video game.	2.2 D
2.3. Software	C. The physical parts that make up a computer.	2.3 E
2.4. Algorithm	D. Picks up the pen to stop drawing.	2.4 A
2.5. Hardware	E. A set of instructions that tells a computer what to do.	2.5 C

Total [5]**Section C – Use information (Apply) (CAPS Pillar)****Question 3:** Read the scenario below and answer the following questions.

You are a young engineer for Boxing Basics, a company that builds video advertisement boards. You are tasked with demonstrating how a basic advertisement board works using a Micro:bit.

3.1. Describe the function of the show icon block: (2) **It displays different icons on the MicroBit's LED array.**

3.2. Explain the difference between the **show icon** and the **show LED** blocks. (2)

Show icon block	Show LED block
Shows a specific set of icons you can choose from.	Allows you to change which LEDs are turned on to make your own icons.

3.3. On your Micro:bit there needs to appear a heart icon and it needs to change into the number 120. How would the number appear on the MicroBit's LED array, given that there isn't enough space for it? (2) **The number will be displayed by moving it across the LED array from right to left.**

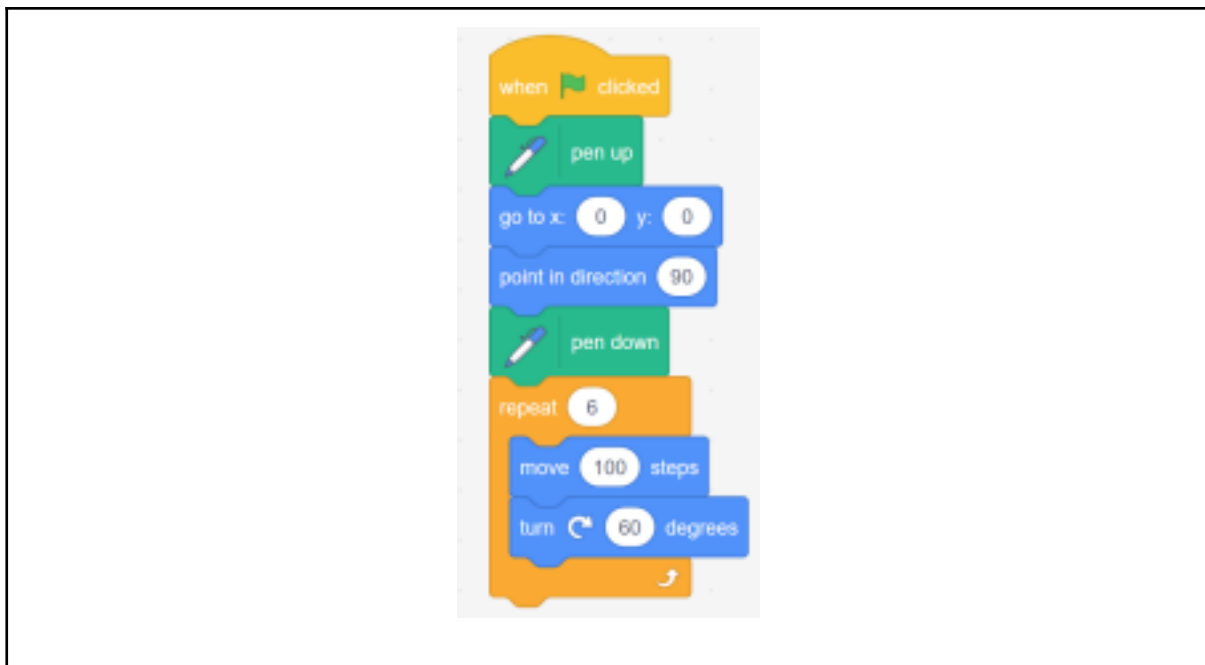
3.4 Identify the difference between the **Wait** block in Scratch and the **Pause** block in MakeCode. (2) **The two blocks have the same function in both; the only difference is the name of the block.**

3.5 When you click on the **Pause** block to change its time to 1 second, it displays the number 1000. Explain why this happens. (3) **The number displayed is in milliseconds, so 1 second equals 1000 milliseconds. Total [11]**



Section D – Draw ideas (Analyse) (CAPS Pillar)

Question 4: Examine the example code below and answer the following questions.



4.1. Explain in full what your sprite does in the following code and what shape will appear at the end. (4)

The sprite lifts up its pen and moves to a specific position and angle. It then puts the pen down to draw and draws six lines that together form a hexagonal shape.

4.2. Will the program still draw a complete hexagon if we change the degrees to 40? Explain your answer in full. (2)

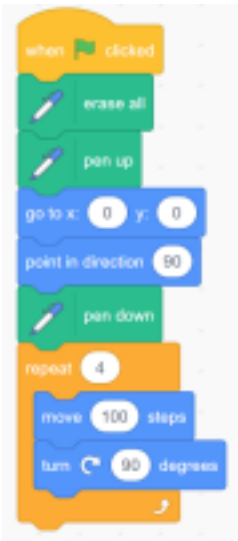
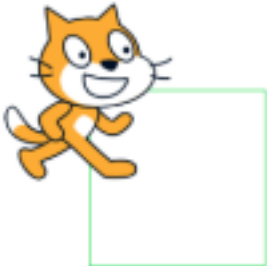
No, the shape will not be complete as the hexagon has 60-degree angles. It will only draw six connected lines that don't form a shape.



4.3. What could you change in the code to make the sprite draw an octagon (a shape with 8 sides)? Identify the blocks you have to change and also the new values. (4)

The repeat block and turn degrees blocks' values need to be changed. You will change the repeat block to a value of 8 and the turn degrees block value to 45 degrees.

You want to create a pattern of squares that overlap using Scratch. The colour of the squares should change every time as well. You are already using the following code, the result and the pattern we want to create.

Code:	Result:	The pattern to create:
		

4.4. Name all the blocks you need to add to make the pattern. (4)

“Repeat” block, “Turn degrees” block and “change pen colour by” block. Total

[14]



Question 5:

Scenario:

Jenny is trying to write an algorithm for a robot designed to replace broken light bulbs in her house.

The robot must check if a light bulb is broken, if it is broken it should remove the bulb and replace it with a new one. Any bulb that isn't broken should be left alone. After the program has run the robot should dispose of the broken bulbs and return to its charging station.

Create an algorithm for the robot's instructions.

Your algorithm should have the following:

- Number each step of the algorithm
- Make sure the steps are in the correct order.
 - Indicate where the robot has to make choices clearly.
- Show all the steps the robot must follow.
- Write neatly and legibly.

1. The robot moves to the first light bulb from the charging station.

2. Check if the light bulb is working.

3. IF lightbulb broken

a. Remove the broken light bulb

b. Insert a new light bulb

4. IF the light bulb is not broken or has been replaced, move to the next light bulb.

5. Repeat steps 1 – 4 for all the remaining bulbs.

6. Dispose of broken bulbs.

7. Return to the charging station. Total [10] Total Marks: 45



RUBRIC: Algorithm

Criteria (2 marks each)	Excellent (2 marks)	Satisfactory (1 mark)	Needs Support (0 marks)
1. Algorithm steps clearly numbered	All steps are clearly numbered and correct.	Some steps are wrong or numbered wrong.	Steps are missing or mostly incorrect.
2. Logical order of steps	Steps follow a clear and logical order.	Some follow a logical order.	The steps are not in logical order.
3. Representation of Key Steps	Includes all key steps.	One or two key steps are missing.	Several key steps are missing or incorrect.
4. Understanding of Concept	Demonstrates a clear understanding of algorithms.	Shows some understanding of algorithms.	Limited understanding of algorithms.
5. Presentation and Clarity	The algorithm steps are neat, easy to read, and clearly numbered with correct use of space.	Mostly neat and readable, but lacks clear numbering or spacing.	Untidy, incomplete, or very difficult to read.